CLASS 9 — MATHS · FORMULA SHEET

Chapter 5: **Circles** | Quick revision, one page

Chords

- The perpendicular from the centre to a chord bisects the chord; the line from the centre to the midpoint of a chord is perpendicular to it.
- Equal chords are equidistant from the centre, and chords equidistant from the centre are equal. Equal chords subtend equal angles at the centre.
- Chord length: $2\sqrt{r^2 - d^2}$ where r is the radius and d the distance from the centre.
- The diameter is the longest chord.

Angles in a circle

- Angle at the centre $= 2\times$ angle at any point of the remaining circle (same arc).
- Angles in the same segment are equal.
- Angle in a semicircle $= 90^\circ$.

Cyclic quadrilateral

- Opposite angles are supplementary: $\angle A + \angle C = 180^\circ$, $\angle B + \angle D = 180^\circ$.
- Exterior angle $=$ interior opposite angle.
- A parallelogram inscribed in a circle is a rectangle.

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